Gabrielle Johnson’s Week 6 Final Deliverable

Dataset Summary

The dataset contains 55,121 emergency department patient encounters across 225 columns, approximately 200 of which are chief complaint columns. The remaining columns were grouped by clinical vitals, demographic features, administrative features, and the ESI levels. Potential leakage variables, such as disposition, were kept separate because they occur after triage and should not be used for prediction.

The dataset is adult only, with approximately 32% being over the age of 65. In terms of demographics, women are the majority of the patients recorded at 58%, additionally, 80% of patients are non-hispanic with 53% being of the White/Caucasian race.

Triage acuity was recorded using the Emergency Severity Index (ESI), which ranges from 1 to 5, where 1 represents the most urgent cases and 5 represents the least urgent cases. Nearly half (49%) of all encounters were classified as ESI 3, while fewer than 1% were classified as ESI 1.

The dataset also includes routinely collected triage vital signs, including heart rate, systolic blood pressure, diastolic blood pressure, respiratory rate, oxygen saturation, temperature, and glucose. Vital signs were reviewed using both plausible clinical limits and normal reference ranges. 29 patients had clinically implausible vital signs which were flagged as possible data quality issues.

The most common chief complaint was abdominal pain, representing 12.8% of encounters, followed by “other” at 7.8%. Although no demographic variables, vital signs, or ESI levels were missing, 52 patients had no recorded chief complaint. These cases were reviewed separately because the absence of a chief complaint may reflect documentation gaps rather than true absence of symptoms.

Model Description

Logistic Regression

The logistic regression model was trained using standardised data because the features were measured on very different scales. For example, heart rate and glucose contain much larger values, while the chief complaint variables are recorded only as 0 or 1. The scaler was fitted using the training data only and then applied to the test data to avoid data leakage. The model was also set to run for more iterations to reduce convergence problems, and a fixed random state of 42 was used so the same results could be reproduced.

The model achieved an accuracy of 0.667 and a weighted F1-score of 0.661. However, the macro F1-score was lower at 0.493, showing that performance was uneven across the ESI categories. The model performed better on the more common ESI 2, 3, and 4 cases, but struggled with the rare ESI 1 cases. It correctly identified only 4 of the 16 ESI 1 patients, giving a recall of 0.250.

Decision Tree

The first decision tree was trained without any limits and performed perfectly on the training data, with an accuracy of 1.000. However, its accuracy on the test set dropped to 0.558, showing that it had overfitted the training data and did not generalise well to new cases.

To reduce overfitting, the tree was retrained with a maximum depth and at least 20 patients in each leaf. Different depth values were tested, and depths of 10, 20, and 30 produced test accuracies of 0.577, 0.601, and 0.614 respectively. A maximum depth of 30 was selected because it gave the best performance.

The final decision tree achieved a test accuracy of 0.614. Although this was an improvement over the unrestricted tree, it still performed worse than logistic regression. It also struggled most with ESI 1 patients. Overall, logistic regression provided the stronger baseline model for this dataset.

Benchmark Table

	ESI1 Recall	ESI1 Precision	ESI2 Recall	ESI2 Precision	ESI3 Recall	ESI3 Precision	ESI4 Recall	ESI4 Precision	ESI5 Recall	ESI5 Precision	Accuracy	F1 Weighted	F1 Macro
Dummy	0.000	0.000	0.327	0.334	0.498	0.491	0.148	0.147	0.045	0.049	0.375	0.375	0.204
Logistic Regression	0.250	0.444	0.609	0.716	0.757	0.661	0.589	0.610	0.111	0.482	0.667	0.661	0.493
Decision Tree	0.000	0.000	0.506	0.710	0.833	0.584	0.247	0.612	0.070	0.378	0.614	0.588	0.349
Logistic Regression (balanced)	0.625	0.017	0.602	0.658	0.500	0.762	0.655	0.488	0.691	0.139	0.563	0.596	0.411
Decision Tree (balanced)	0.500	0.054	0.604	0.617	0.433	0.699	0.582	0.380	0.588	0.110	0.516	0.539	0.378

Metric Justifications

The primary evaluation metric for this project is per class recall across ESI levels 2–4, with ESI 1 recall monitored separately. Although ESI 1 represents the highest clinical priority because these patients require immediate life saving intervention, such cases are often clinically apparent at presentation. The model's greatest value lies in the more ambiguous middle categories, such as distinguishing an emergent ESI 2 patient from a stable ESI 3 patient. Recall is prioritised because it directly measures how many patients within each ESI category are correctly identified and, consequently, how many cases are missed. Accuracy cannot capture this concern on an imbalanced dataset because 49% of encounters are ESI 3 and fewer than 1% are ESI 1, a model can score above 60% accuracy while failing to correctly prioritise a single critical patient. Therefore, reporting recall separately for each class prevents the ESI 3 category from dominating and masking poor performance on less common .Precision and F1-score will still be reported alongside recall to ensure that improvements in identifying urgent patients do not result in excessive over triage/false alarms.

Failure Mode Reflection

The model performs worst on ESI 1, with the baseline logistic regression correctly identifying only 4 of 16 critical patients. A likely contributing factor is class scarcity, as ESI 1 accounts for fewer than 1% of encounters, leaving the model with very few examples from which to learn.

By volume, the more significant failure occurs in ESI 2, with a recall of 0.61 meaning that 1,399 of 3,585 emergent patients were missed, with 1,343 classified as ESI 3 instead. This could delay care for patients with serious but less obvious or time sensitive presentations. ESI 5 errors are mainly “safer’, with 213 of 243 patients being over triaged. Although this is less life threatening than under triage, it could increase unnecessary resource use and produce false alarm.

Applying class weighting improved ESI 1 recall to 0.625 but reduced precision to 0.017, producing approximately 59 false positive ESI 1 predictions for every correct identification. This trade off suggests that class weighting alone is not a suitable solution.

Next Step

Building on these results, the next phase will test whether stronger models can improve on the logistic regression baseline. XGBoost and CatBoost will be compared because similar models performed well in the reviewed studies and may be better at recognising more complex patient patterns. The next stage will also focus on reducing under triage, especially for ESI 1 and ESI 2 patients, while avoiding too many false alarms.

Below is a link to my recorded clinical justification:
https://www.loom.com/share/837c2b7f744946b492ceacdd12e4b9db

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